

Problem Framing and Research Question

Passenger Demand forecasting helps transportation become more efficient through pinpointing where to allocate more buses and where to reduce some. In line with this, this investigation hopes to identify how average boarding is related with locations, day of the week, time period, part of the year, and average alightings.

Meta	Features	Target
Route Number and Stop Number	- Coordinate X - Coordinate Y - Schedule Period - Day Type - Time Code - Average Alightings	Average Boarding

Table 1. Role assignment for each variable.

Each features determine the relationship of average boarding in relative to:

- **x-coordinate:** east-west location
- **y-coordinate:** north-south location
- **Schedule Period:** different periods of the year
- **Day Type:** weekdays, Saturdays, and Sundays
- **Time Code:** time of day
- **Average Alightings:** the number of passengers getting off

Workflow Reproducibility Audit

DSML Tool Identification

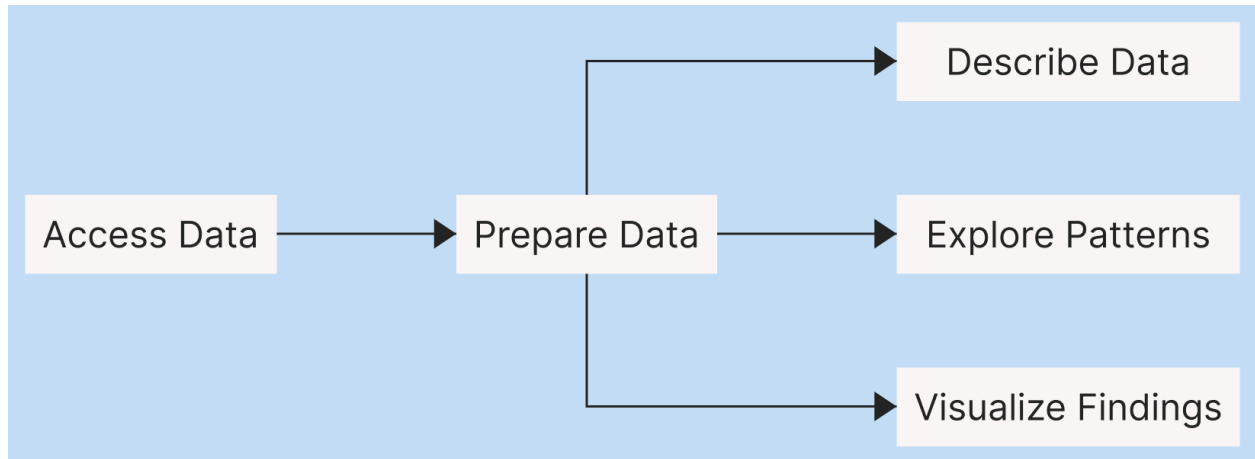


Figure 1. Pipeline Construction

- **Access Data**
 - CSV File - Loads the dataset into the Orange workflow
- **Prepare Data**
 - Select Columns - Filter out unwanted variables and assign their roles
- **Describe Data**
 - Feature Statistics - Summary about the variable
- **Explore Patterns**
 - Distributions - shows how values are distributed
- **Visualize Findings**
 - Scatter Plot - shows relationship with numerics
 - Bar Plot - shows relationship with categories

Local vs. Cloud Environments

This investigation runs Orange locally as it provides quick access to the dataset without relying on internet or cloud services. But performance heavily relies on the computer's resources. In contrast, the cloud offers more resources but heavily relies on the internet.

Visual Mining vs. Code

This investigation uses visual mining for easier and quickly connecting data preparation, exploration, and visualization through widgets in order to understand patterns and relationships before using Python for more complex analysis.

Data Exploration and Summary Findings

For this exercise, the data used is retrieved from Winnipeg Transit's Open Data for Estimated Boarding starting from June 2025. The dataset has 186,980 observations with no missing values. Using Feature Statistics, the researcher found that the average boardings has a mean of 5.915, a minimum of 0, a maximum of 969.8, and a dispersion of 2.896. On the other hand, average alightings has a mean of 6.158, a minimum of 0, a maximum of 917.9, and a dispersion of 2.958. Average alightings have a higher dispersion while average boardings have a larger range.



Figure 2. Screenshot of feature statistic widget.

Assignment #2

The Distributions widget shows that both average Boardings and average alightings are strongly right-skewed as 93.2% of observations for the average boardings are below 20, while 92.9% of observations for average alightings are below 20. In addition, there are unusually high values appearing in average boardings, for example, two average boardings fall between 960 and 980. The same is true for average alightings, two of the values fall between 900 and 920. For the schedule period, the distribution is relatively close, ranging from 24-27% which is totally different from time code where each category ranges from 10.67% to 26.51% Day type, however, contains 55.32% for weekday, 23.13% for saturday, and 21.55% for sunday.

Pattern and Relationship Analysis

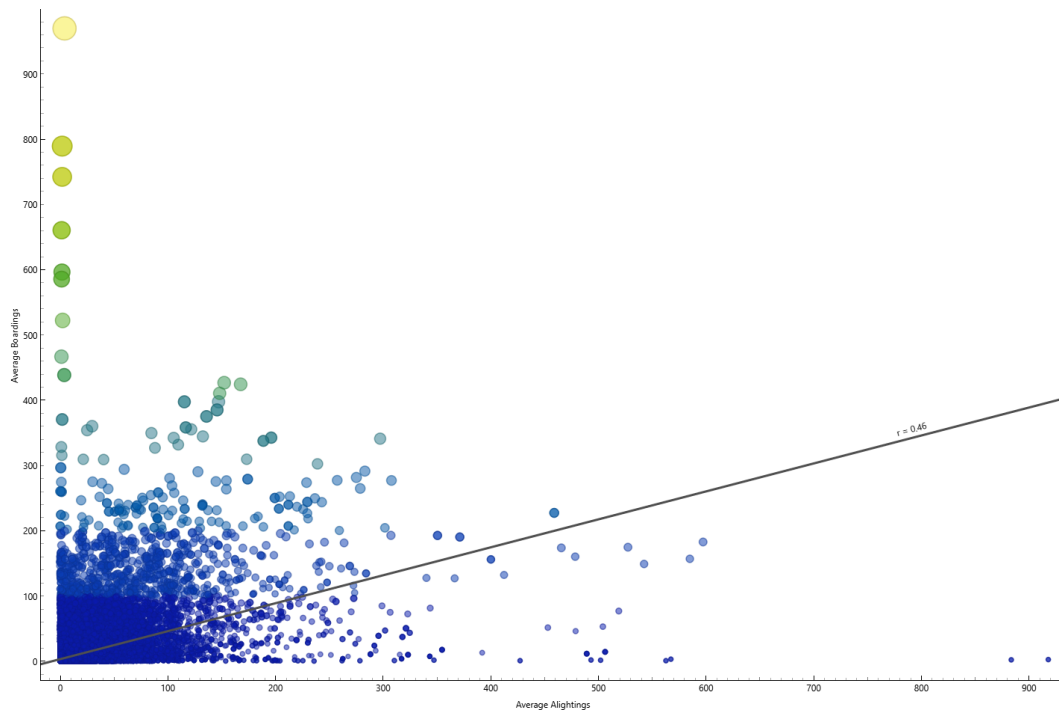


Figure 3. Scatter plot for Average Alightings and Average Boardings.

Assignment #2

The relationship between average alighting and average boarding is quite concentrated near the origin because of their low values. However, with the regression line enabled, Scatter Plot shows that the two of them have a moderate positive relationship ($r = 0.46$). With the restrictions of maximum 200 entries, the schedule period contains only one category (2026 Q1). In addition, most categories in time code are slightly right-skewed with an exception to Ch which has two peaks on the edge. Finally, weekdays have much more scattered values but their shape is close to right-skewed. Same with weekdays, Saturday, though little, is close to a right-skewed distribution. Sunday, however, has almost no distribution.

Conclusions and Analytical Boundaries

The analysis shows that average alightings has a moderate positive relationship with average boardings ($r = 0.46$), while time, day, and location may also help explain passenger demand. However, these patterns show correlation, not causation, and other factors such as weather or major events are not included.



RRC Polytech

Passenger-Demand Forecasting for Winnipeg Transit using LSTM

File:

Becaro_Jovan_Assignment1_Workflow.ows

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